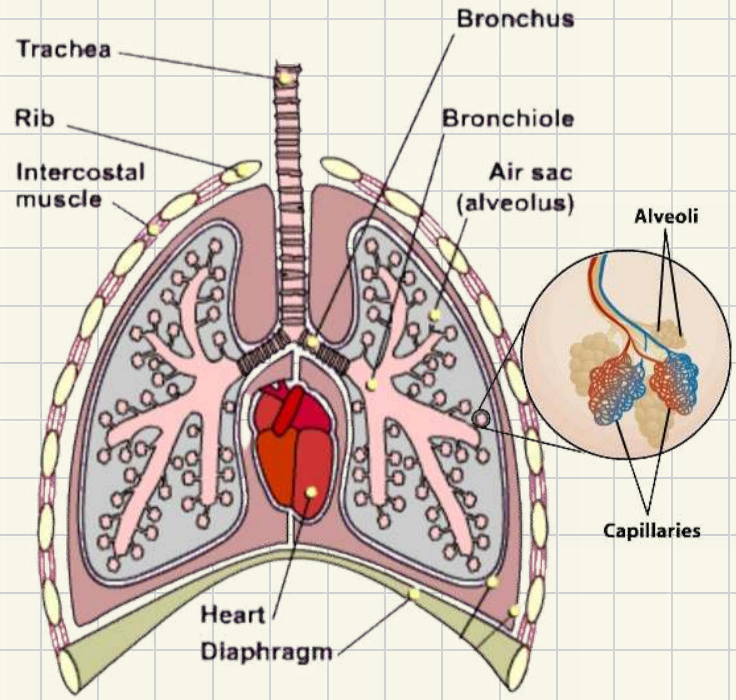
19/01/25

gas exchange

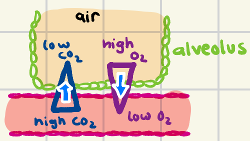


Function of the gas exchange system:

i) filtering the dust & pathogens in the air

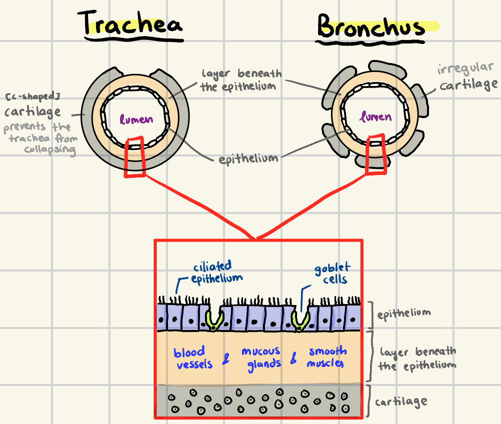
if they enter the lungs, the alveoli can get damaged. These dust & pathogens are filtered out as they pass the trachea

ii) minimize the distance of diffusion between air & blood



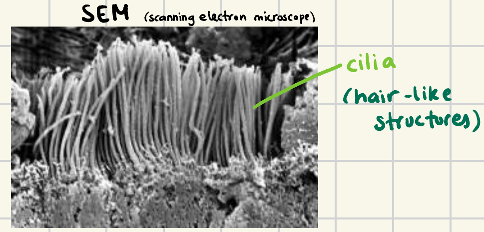
* gas exchange must happen all the time!

iii) maintain the gradient between air in alveoli & the blood in capillaries

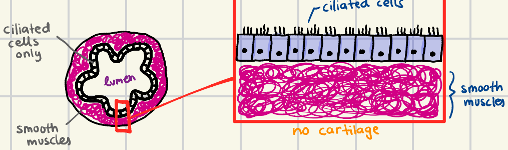


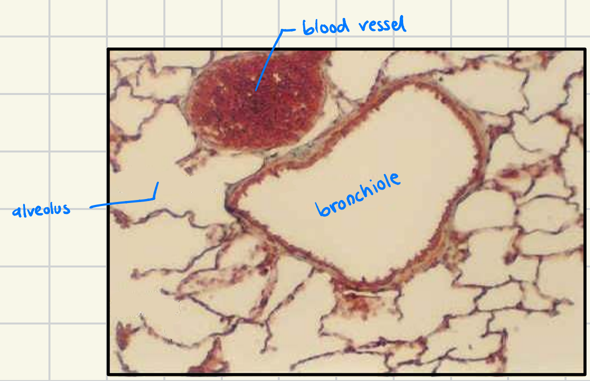
light microscope

SEM (scanning electron microscope)

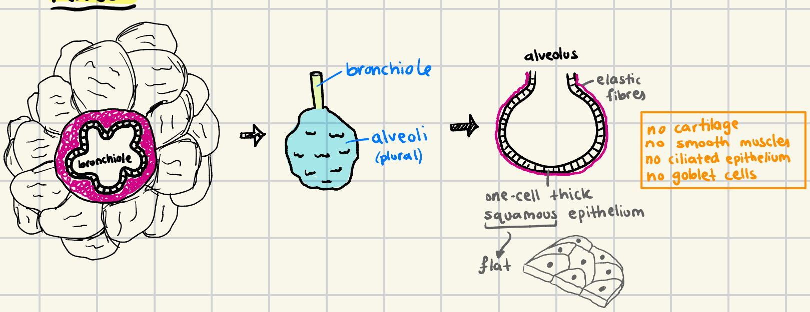


Bronchiole



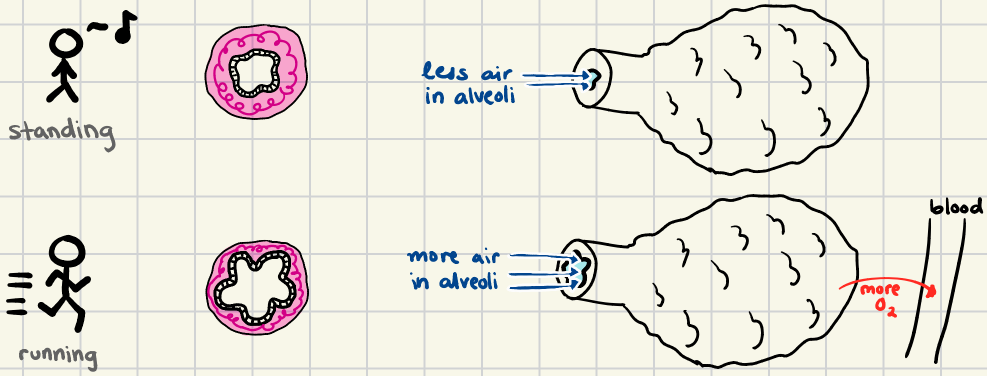


Alveoli

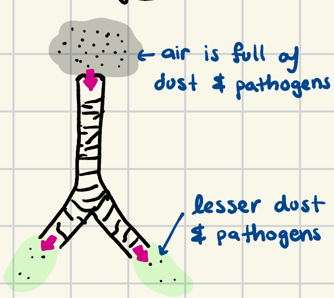


Summary	trachea	bronchus	bronchioles	alveoli
epithelium	- ciliated epithelium - goblet cells		- ciliated epithelium	- squamous epithelium
beneath epithelium	blood vessels & mucous glands & smooth muscles		blood vessels smooth muscles	elastic fibres
cartilage	c-shaped	irregular	X	X

Response of bronchioles during exercise

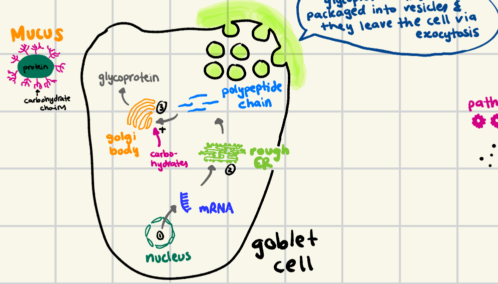


Filtering of dust & pathogens

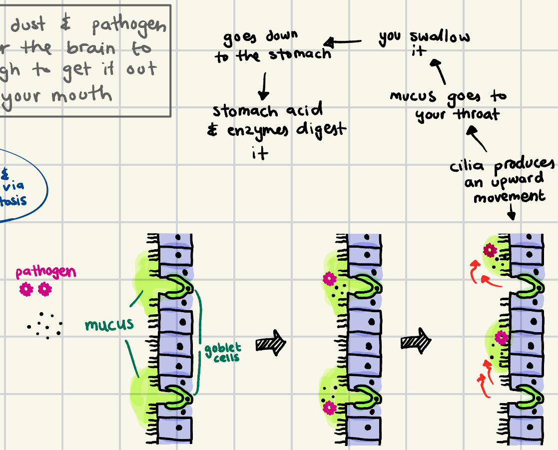


HOW?

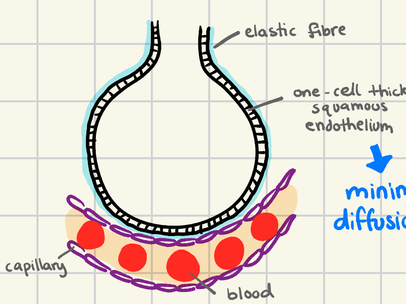
⇒ goblet cells



too much dust & pathogen will trigger the brain to sneeze/cough to get it out through your mouth

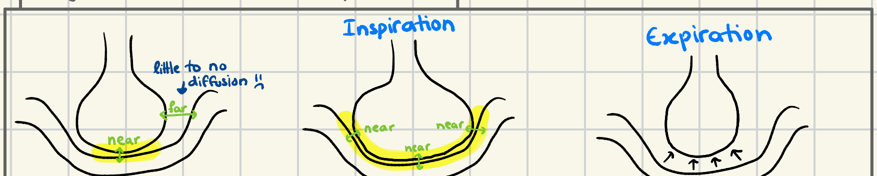


Minimizing diffusion distance



minimizes the diffusion distance

why is the elastic fibre important ?

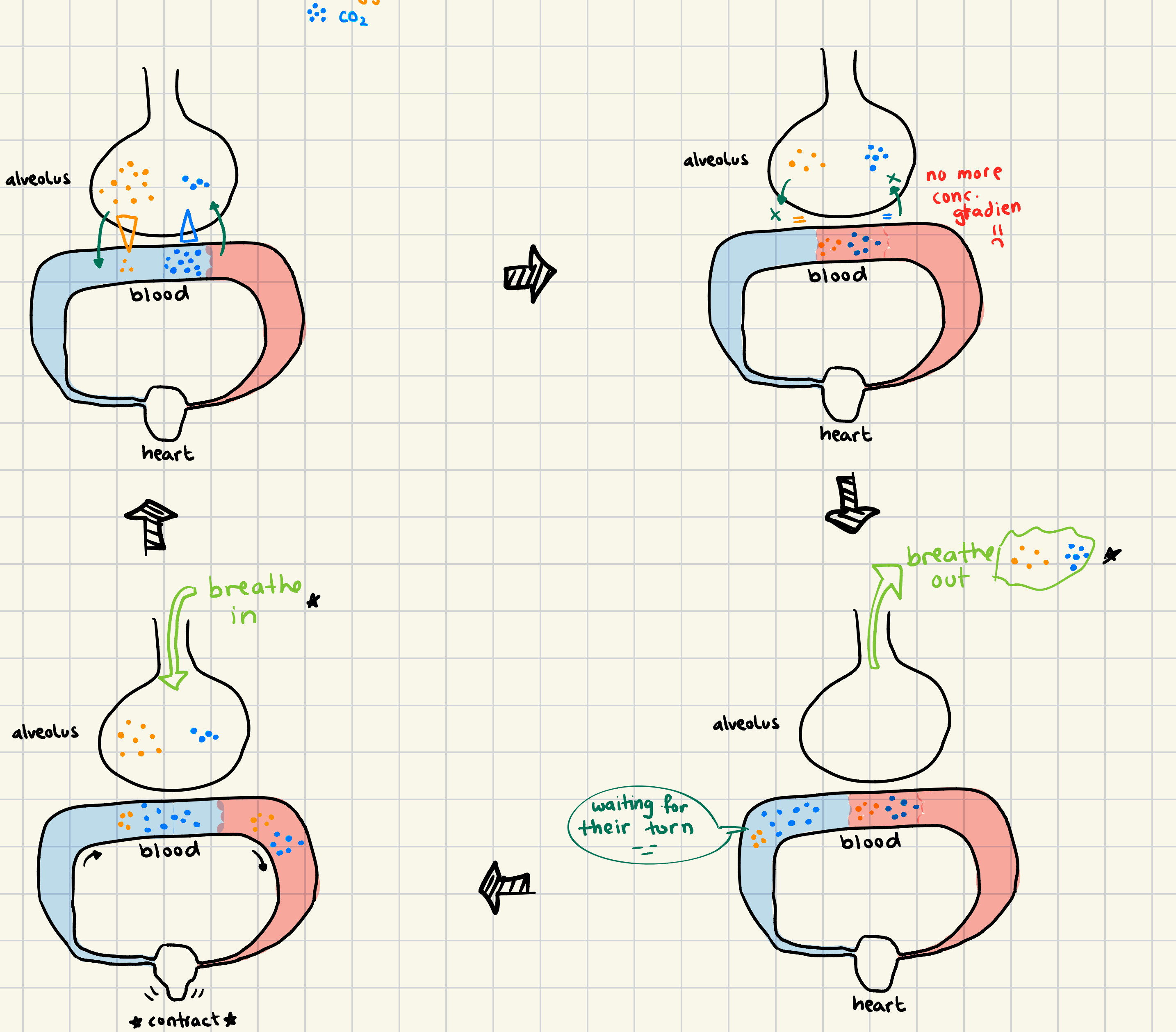


minimum diffusion w/o elastic fibre

elastic fibre allows alveolus to stretch & thus provides more area for diffusion to occur

elastic fibre allows alveolus wall to recoil to expel the air out easily

Maintaining the conc. gradient



★ To maintain the conc. gradient :

① lung ventilation ensures that the alveolus will always have a high PPO_2 & low $PPCO_2$

② heart contraction pushes deoxygenated blood to lungs, ensuring that blood has a low PPO_2 & a high $PPCO_2$